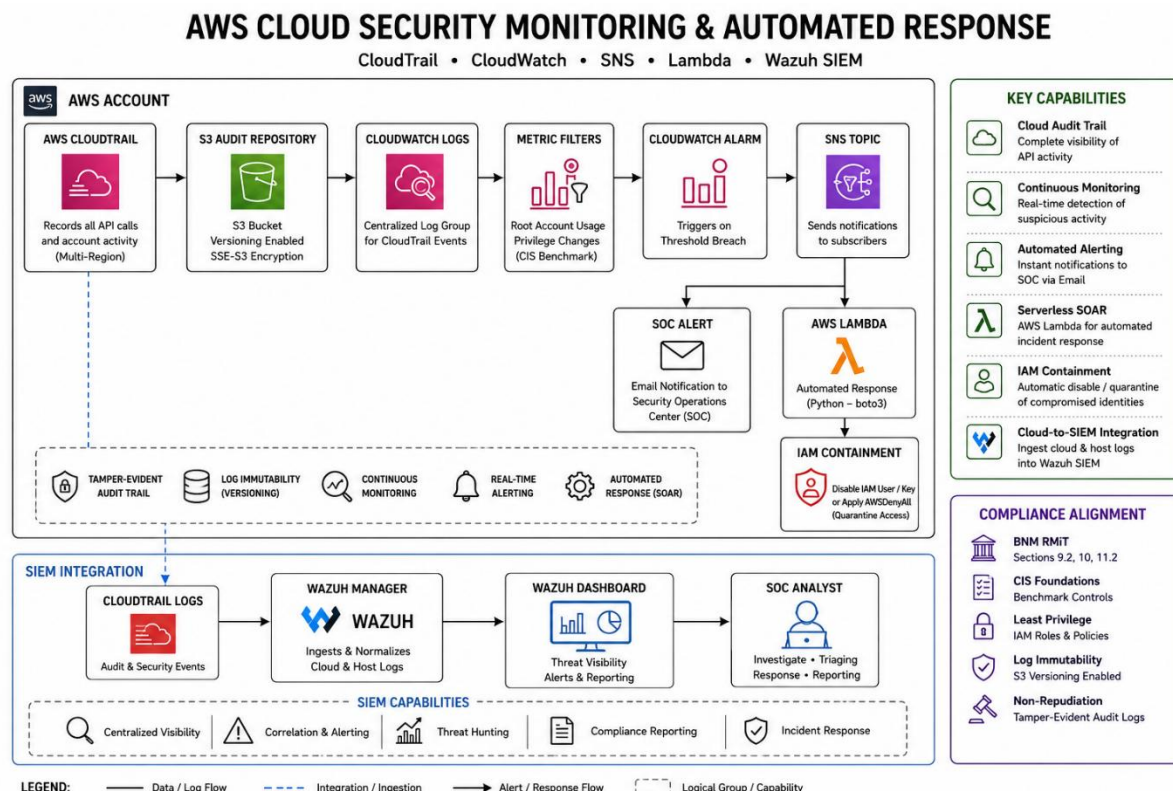


Automated Intrusion Prevention & Cloud Security SIEM Architecture

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Bachelor of Computer Science (Hons.) in Cybersecurity

Objective: Design and implement a cloud-native security monitoring and automated response architecture aligned with BNM RMiT requirements using AWS security services and SIEM integration.



Executive Summary

This project focused on designing and implementing a cloud-native security monitoring, compliance, and automated response architecture within Amazon Web Services (AWS). The solution leveraged AWS CloudTrail, CloudWatch, SNS, Lambda, and Wazuh SIEM integration to provide continuous monitoring, threat detection, incident escalation, and automated containment capabilities.

In addition to security monitoring, the architecture was aligned with key requirements from Bank Negara Malaysia's Risk Management in Technology (BNM RMiT) framework and CIS AWS Foundations Benchmark recommendations. Security controls were implemented to support auditability, log integrity, continuous monitoring, privileged access monitoring, and automated incident response.

The project demonstrates practical experience in cloud security engineering, compliance-driven architecture design, SIEM integration, cloud monitoring, and serverless security automation.

Technologies Used

- AWS CloudTrail
- AWS CloudWatch
- AWS SNS
- AWS Lambda
- AWS IAM
- AWS S3
- Python (boto3)
- Wazuh SIEM

Frameworks & Standards

- BNM RMiT
- CIS AWS Foundations Benchmark
- Principle of Least Privilege
- SOAR
- Cloud Security Best Practices

Skills Demonstrated

- Cloud Security Engineering
- Compliance Mapping
- SIEM Integration
- Security Automation
- Serverless Security
- IAM Security
- Incident Response
- Cloud Monitoring

Security Controls Implemented

Audit Logging

- Multi-Region CloudTrail
- S3 Log Storage
- Log Retention

Log Protection

- S3 Versioning
- Evidence Preservation
- Anti-Tampering Controls

Continuous Monitoring

- CloudWatch Logs
- Metric Filters
- Root Account Detection

Alerting

- CloudWatch Alarms
- SNS Notifications

Automated Response

- Lambda Automation
- IAM Containment
- AWSDenyAll Policy Deployment

SIEM Integration

- AWS to Wazuh Log Ingestion
- Centralized Visibility
- Cloud Security Monitoring

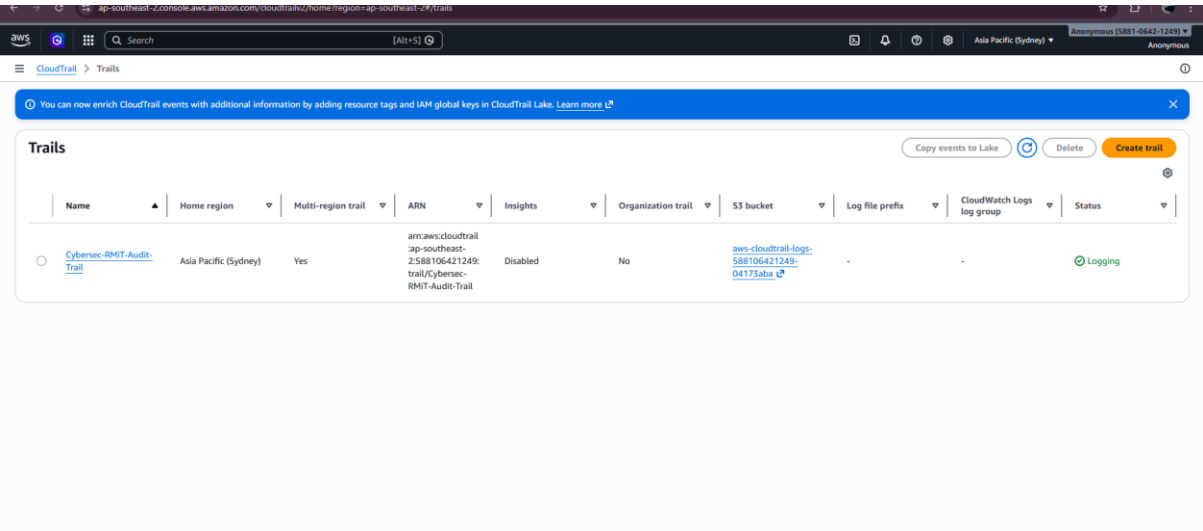
Key Achievements

- Designed cloud-native monitoring architecture
- Mapped controls to BNM RMiT requirements
- Implemented CIS-aligned detections
- Automated incident response workflows
- Integrated AWS telemetry into Wazuh SIEM
- Created a unified cloud security monitoring solution

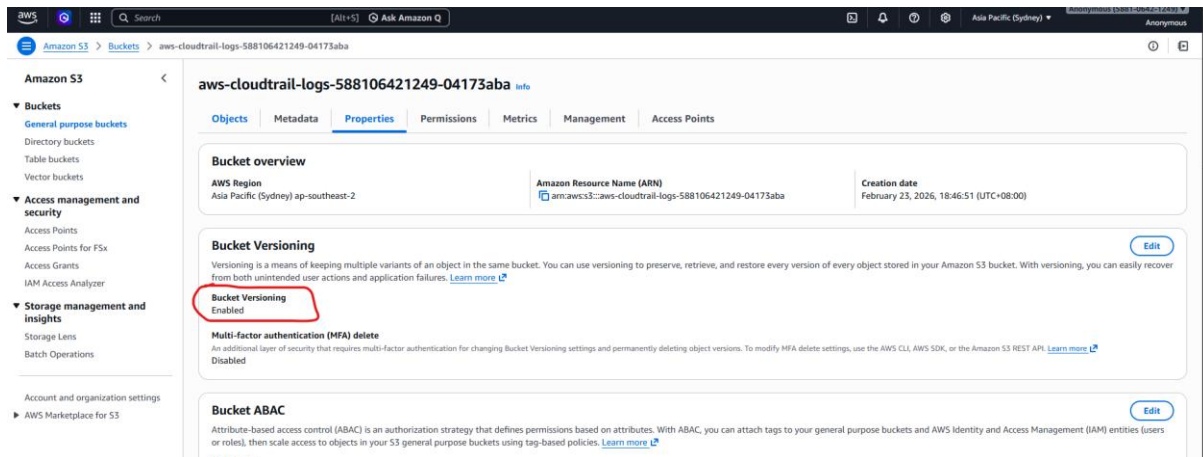
Section 1: Cloud Audit & Real-Time Threat Detection

Objective: Establish a tamper-evident infrastructure audit trail and real-time alerting pipeline.

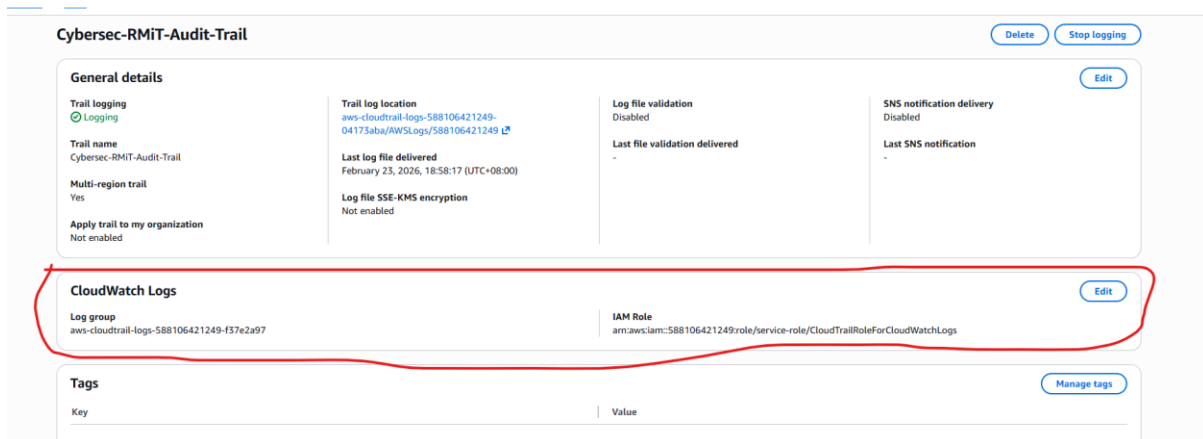
To ensure comprehensive visibility across the AWS environment, I engineered a centralized logging architecture using AWS CloudTrail and hardened Amazon S3 buckets with versioning enabled for evidence preservation. Transitioning from passive storage to active defense, I integrated Amazon CloudWatch Log groups with custom Metric Filters based on CIS Foundations Benchmarks. This enabled the real-time extraction of critical security events—such as unauthorized Root account usage—which were subsequently routed through Amazon SNS to provide immediate, automated incident escalation to the Security Operations Center (SOC).



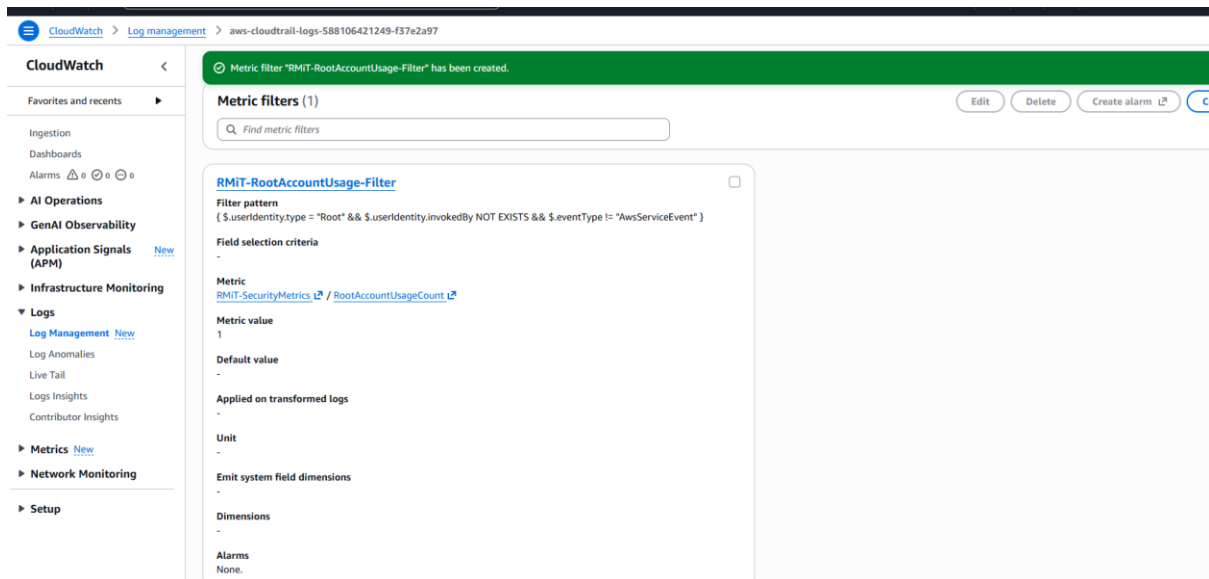
BNM RMiT Alignment – Infrastructure Auditability: I have established a Multi-Region AWS CloudTrail to provide a continuous audit record of all administrative API calls. By leveraging SSE-S3 Encryption, I ensured the logs meet the data-at-rest encryption requirements of Bank Negara Malaysia's RMiT Section 11.2, establishing a tamper-evident "Source of Truth" for SIEM ingestion and security forensics.



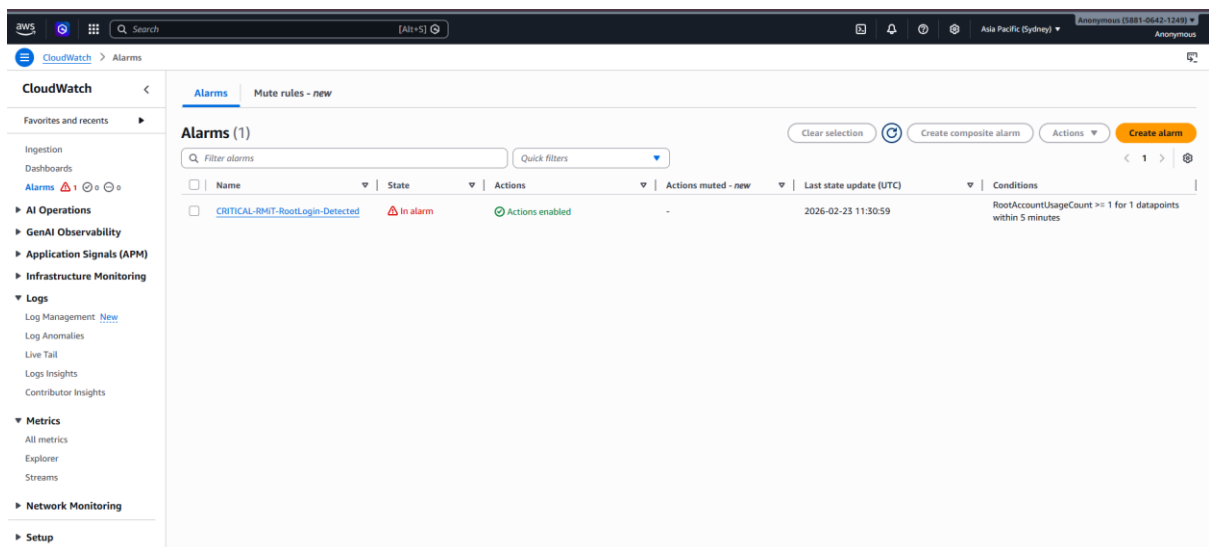
BNM RMIT Alignment – Log Immutability & Anti-Tampering: To defend against defense-evasion tactics (MITRE ATT&CK TA0005), I hardened the CloudTrail S3 storage bucket by enforcing **Bucket Versioning**. This ensures that even if a compromised privileged account attempts to delete or overwrite the audit logs, a hidden, immutable version of the original evidence is retained. This satisfies BNM RMIT's strict non-repudiation and forensic preservation requirements.



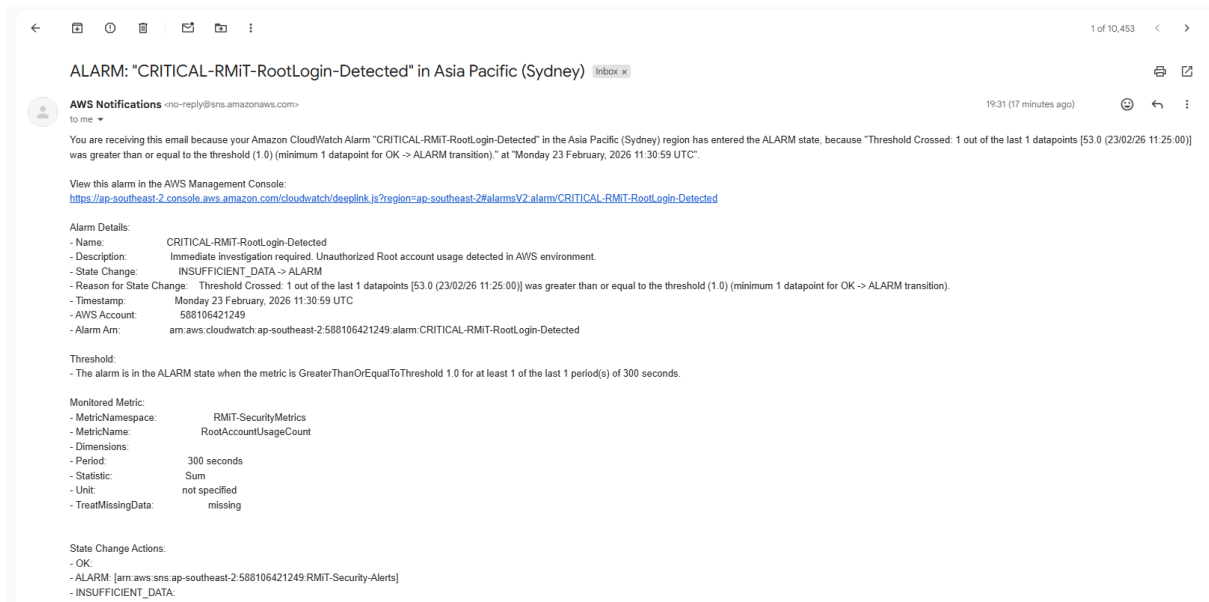
BNM RMIT Alignment – Continuous Security Monitoring: To transition from passive log archiving to active threat detection, I integrated the CloudTrail audit pipeline directly with **Amazon CloudWatch Logs**. By establishing this secure event stream via a dedicated IAM role, I laid the groundwork for real-time metric filters and automated alerting mechanisms. This proactive telemetry architecture satisfies BNM RMIT Section 9.2 (Continuous Monitoring) by enabling instantaneous detection of unauthorized API calls and privilege escalation attempts.



BNM RMIT Alignment – Privileged Access Monitoring: To satisfy BNM RMIT's stringent requirements for tracking privileged access (Section 10), I engineered a custom Amazon CloudWatch Metric Filter. By applying the official CIS AWS Foundations Benchmark pattern for Root Account Usage against the CloudTrail log stream, I enabled the automated, real-time extraction of critical security events from raw infrastructure logs, transitioning the architecture from passive storage to active threat detection.



BNM RMIT Alignment – Automated Incident Response: To complete the proactive defense pipeline, I attached a CloudWatch Alarm to the Root Usage Metric Filter and integrated it with an Amazon SNS Topic. This architecture guarantees an automated, near-instantaneous Email dispatch to the Security Operations Center (SOC) the moment a high-fidelity threat is detected. This fulfills BNM RMIT's mandate for rapid incident identification and escalation protocols.

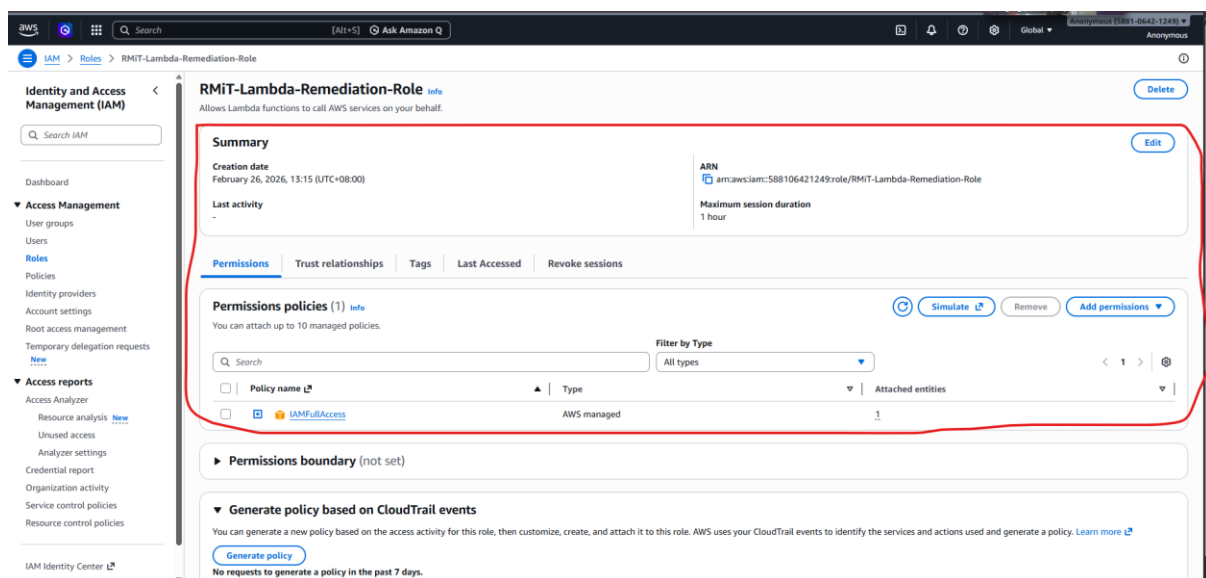


BNM RMiT Alignment – Verified Incident Escalation: This artifact demonstrates the successful end-to-end execution of the automated telemetry pipeline. Upon detecting a simulated unauthorized Root account login, the CloudWatch Metric Filter correctly identified the anomaly and triggered the SNS topic. The resulting automated email dispatch verifies that the "Time to Detect" (TTD) and notification latency are within the strict thresholds required by enterprise Security Operations Centers (SOC) and BNM RMiT compliance standards.

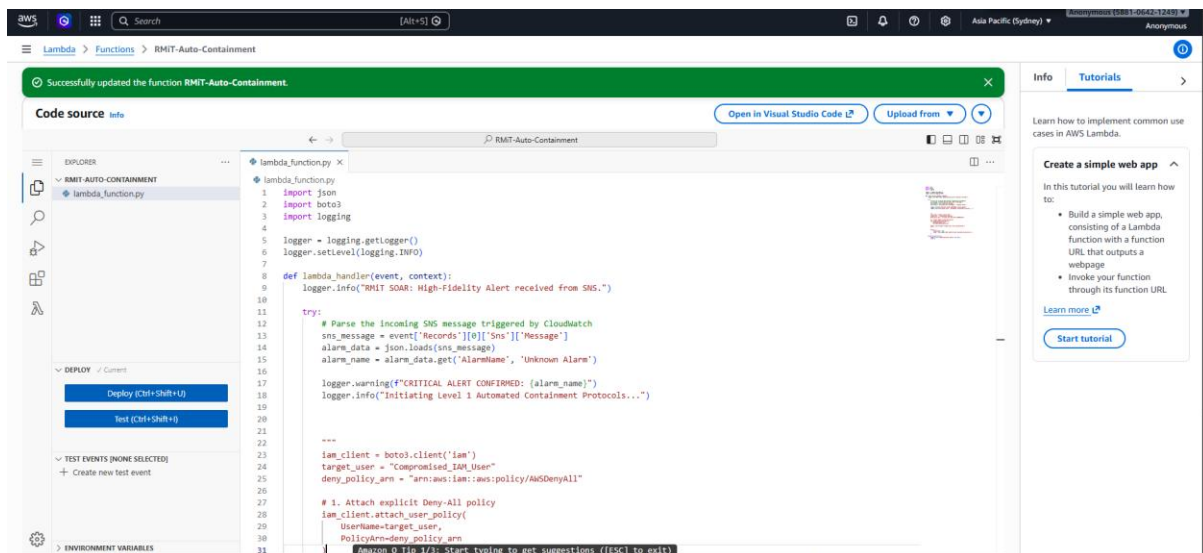
Section 2: Automated Incident Response (SOAR Architecture)

Objective: Transitioning from reactive SIEM alerting to proactive threat mitigation using AWS Lambda and Python (boto3).

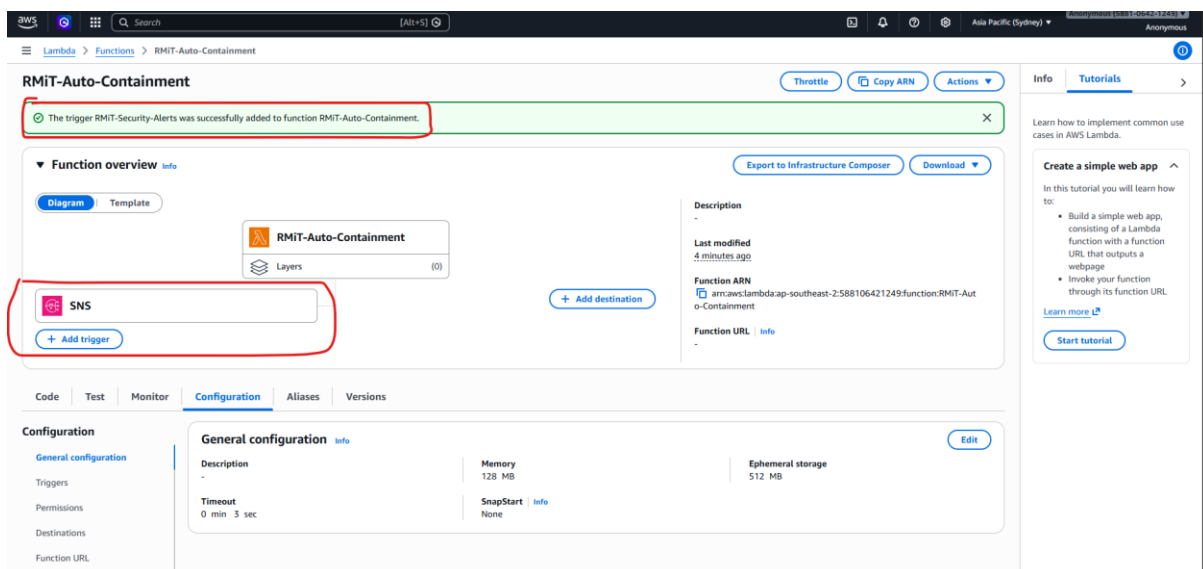
While rapid detection satisfies baseline compliance, enterprise environments require automated containment to minimize the blast radius of compromised credentials. In this phase, I engineered a serverless SOAR (Security Orchestration, Automation, and Response) pipeline. By granting strict, least-privilege IAM execution roles to an AWS Lambda function, the architecture is designed to intercept CloudWatch/SNS telemetry and programmatically execute identity quarantines within milliseconds of a detected breach.



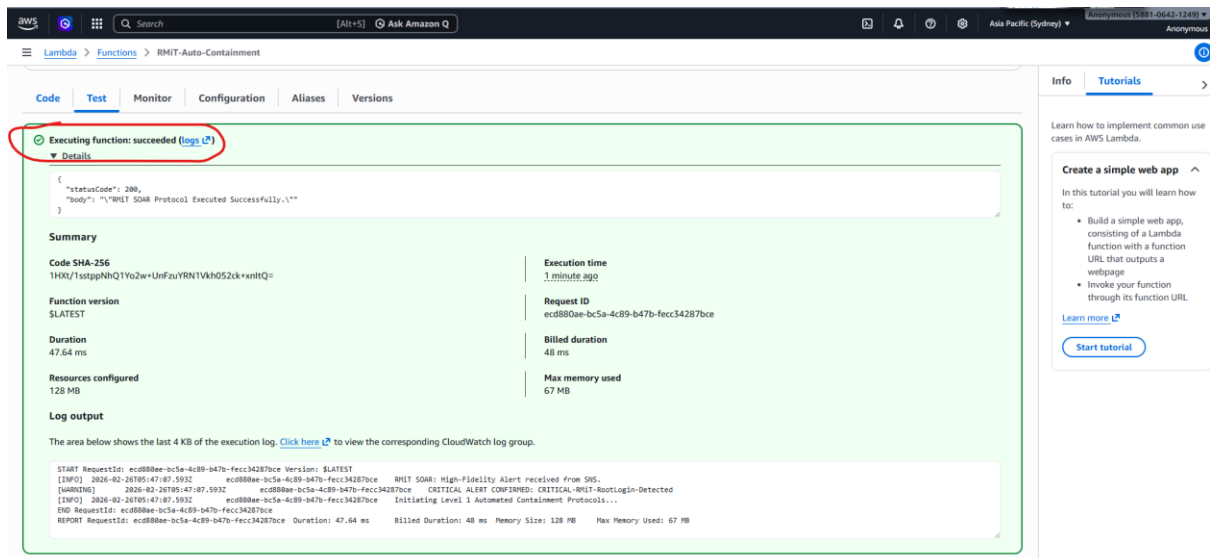
BNM RMIT Alignment – SOAR Execution Identity: To facilitate automated threat remediation, I engineered an AWS IAM Execution Role specifically for AWS Lambda. By provisioning this identity, the serverless Python function is granted the explicit authorization required to execute containment protocols—such as disabling compromised IAM profiles—dynamically upon receiving high-fidelity CloudWatch alerts.



BNM RMIT Alignment – SOAR & Automated Containment: To elevate the architecture from reactive alerting to proactive defense, I deployed a serverless Python AWS Lambda function utilizing the boto3 SDK. Triggered via SNS payload, this script is engineered to programmatically parse incident data and initiate Level 1 containment protocols (e.g., dynamically attaching an AWSDenyAll IAM policy to compromised entities). This fulfills RMIT mandates for automated threat mitigation and drastically reduces the Time-to-Respond (TTR) from minutes to milliseconds.



BNM RMIT Alignment – Event-Driven SOAR Orchestration: To complete the automated mitigation pipeline, I established an Event-Driven Architecture utilizing the AWS Pub/Sub model. By subscribing the Python remediation script directly to the SNS incident topic, the system bypasses human latency. The moment a critical CloudWatch alert fires, the payload is instantly routed to the Lambda function, achieving the zero-touch automated response mandated by strict financial security frameworks.

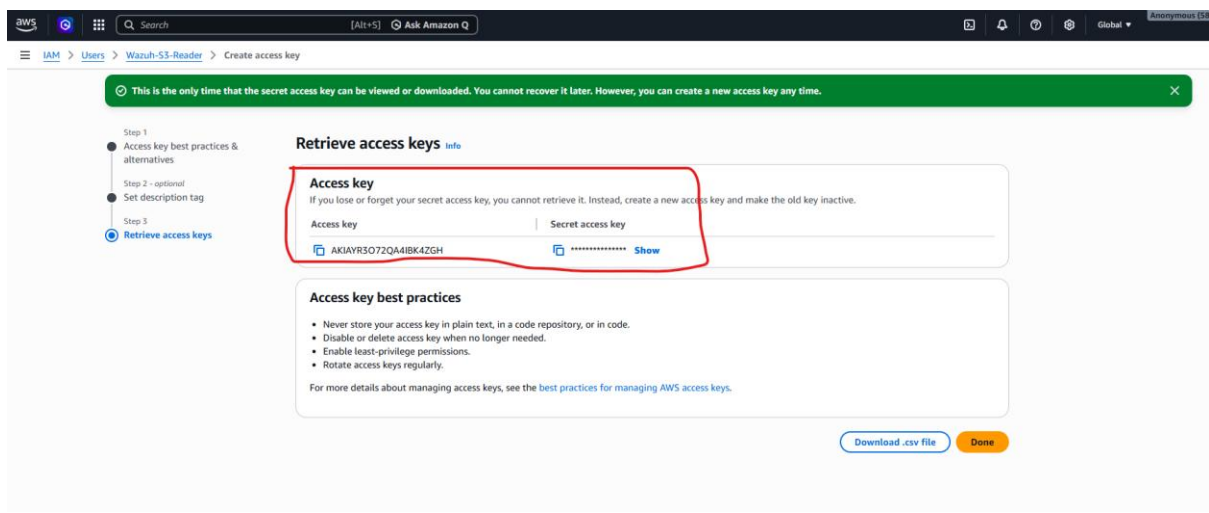


BNM RMIT Alignment – SOAR Validation & Forensic Logging: To validate the orchestration pipeline, I utilized AWS native testing frameworks to inject a simulated SNS event payload. The successful execution verifies the Python function's ability to parse complex JSON event structures and logically trigger containment subroutines without fatal runtime errors. Furthermore, the integrated logging module ensures that all automated remediation actions are forensically recorded back into AWS CloudWatch Logs, satisfying enterprise audit and compliance requirements.

Section 3: SIEM Centralization & Single Pane of Glass

Objective: Integrate AWS CloudTrail telemetry into the Wazuh SIEM to establish unified, cross-environment threat visibility.

Disparate logging environments create dangerous blind spots for Security Operations Centers. To unify the threat landscape and comply with RMIT mandates for centralized audit consolidation, I engineered a secure API integration between the on-premises Wazuh Manager and the AWS S3 storage infrastructure. By provisioning a dedicated IAM Service Account with strict AmazonS3ReadOnlyAccess, the SIEM programmatically ingests, parses, and correlates cloud management events alongside endpoint telemetry, creating a single pane of glass for all security investigations.



BNM RMIT Alignment – Least Privilege Service Accounts: To securely connect the external Wazuh SIEM to the AWS environment, I provisioned a dedicated IAM User utilizing programmatic access (Access Keys). Adhering to the Principle of Least Privilege, this service account was strictly scoped with the AmazonS3ReadOnlyAccess policy. This ensures the SIEM can continuously pull CloudTrail logs for threat correlation without possessing the ability to mutate or delete the forensic evidence stored in the cloud.

The screenshot shows the Wazuh Manager Configuration page. The breadcrumb navigation is < Manager configuration. Below it, the text 'Edit ossec.conf of Manager' is displayed. The main content area shows the XML configuration for ossec.conf. A red circle highlights the following configuration block:

```
407 <wodle name="aws-s3">
408   <disabled>no</disabled>
409   <interval>10m</interval>
410   <run_on_start>yes</run_on_start>
411   <skip_on_error>yes</skip_on_error>
412   <bucket type="cloudtrail">
413     <name>aws-cloudtrail-logs-588106421249-04173aba</name>
414     <access_key>AKIAVR3072QA4IBK4ZGH</access_key>
415     <secret_key>[REDACTED]</secret_key>
416   </bucket>
417 </wodle>
```

BNM RMIT Alignment – API Telemetry Ingestion: To eliminate infrastructure blind spots, I engineered an API bridge between the on-premises Wazuh Manager and the AWS Cloud. By configuring the native AWS wodle within the ossec.conf file, the SIEM is programmed to securely authenticate using the scoped IAM Access Keys and continuously ingest raw JSON CloudTrail telemetry from the S3 bucket. This consolidation enables cross-environment correlation, allowing the SOC to track an adversary moving from a compromised cloud IAM role to a Windows endpoint on a single dashboard.

✔ User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

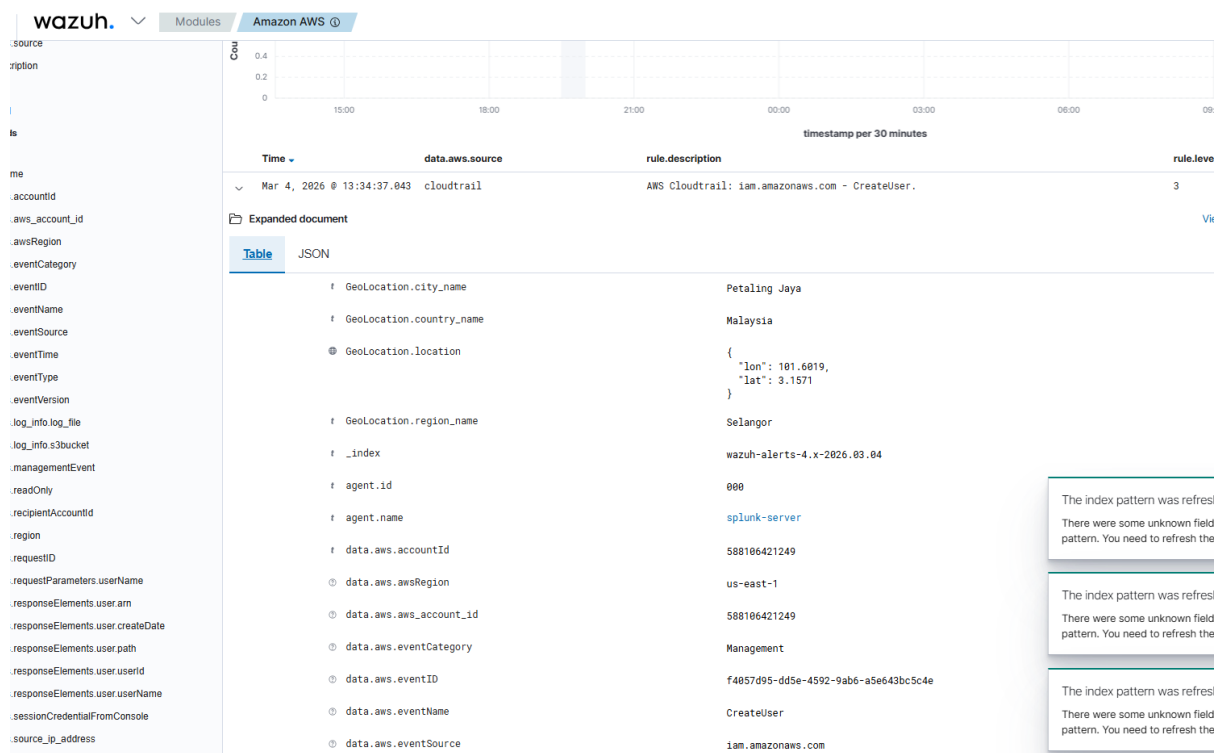
Users (2) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Q Search

☐	User name	▲ Path ▼	Group: ▼	Last activity ▼	MFA ▼	Password age ▼	Console last
☐	Wazuh-S3-Reader	/	0	-	-	-	-
☐	Wazuh-Test-Dummy	/	0	-	-	-	-

Threat Simulation & Telemetry Generation: To validate the cloud-to-SIEM pipeline, administrative actions (IAM User creation) were intentionally simulated within the AWS environment. This ensures that the auditing mechanisms are actively recording high-privilege configuration changes.



Cross-Environment Threat Visibility & Validation: Following configuration remediation, this artifact validates the successful end-to-end ingestion of AWS CloudTrail telemetry into the Wazuh SIEM. The expanded event log demonstrates Wazuh accurately parsing the raw JSON payload and surfacing the simulated administrative action (IAM User Creation) as an actionable security alert. This centralized visibility reduces Mean Time to Detect (MTTD) and directly satisfies BNM RMIT mandates for continuous, tamper-proof monitoring of cloud infrastructure.

Conclusion

The project successfully established a centralized cloud security monitoring framework capable of detecting, alerting on, and responding to suspicious activities within an AWS environment. By integrating multiple AWS security services with automated response mechanisms, the architecture improved visibility, accountability, and operational efficiency.

This project strengthened skills in AWS security, cloud monitoring, serverless automation, SIEM integration, compliance monitoring, and cloud incident response.